Waves On A String PhET Simulation

SYNOPSIS

What are waves, and why do we study waves in physics? Waves can be actual things, such as ocean waves, sound waves, or waves in a slinky when shaken side to side. Waves can also be a description of motion that goes back and forth, such as a pendulum swing or a planet orbiting a sun. The "wave motions" are repetitive, oscillatory, or cyclical; whether they are a thing or description, these cyclical behaviors are ubiquitous (i.e., all around us) and are all mathematically described using sine or cosine functions (or some combination). So, if there are repetitions (whatever it may be, thing, motion, or even stuff), you can bet that any combination of sine or cosine functions will describe this repetition. You will be surprised by the ubiquitousness of wave-like (repetitive) things or behaviors in nature and how cosinusoidal (combining sine and cosine) math functions with only three parameters (amplitude, wavelength, and frequency; more on these later) describe them all.

Why study waves? Because let me repeat: Actually, everything is a wave. You will agree with me when we study Modern Physics. For now, I will say that initially, waves as a thing were thought to be distinct and behave quite differently from things like rocks or cannonballs, in that unlike rocks or cannonballs, which can be thought of as particles (localized things), ocean waves, sound waves, or waves in a string are spread out and do things that particles can never do, such as go through each other, add or subtract from each other. For this Unit, waves are a necessary addition to the particle-centric worldview we are familiar with, such as a flying cannonball whose position and velocity can be pinpointed. Since a wave is spread out (imagine the peaks and troughs of an ocean wave extending into the horizon), we can't pinpoint where the wave is (it's "everywhere") like we can with a particle, thus requiring a different description using cosinusoidal functions. Note that waves can be moving things, just like a flying cannonball is a moving thing, but their repetitive features are spread out in both space and time! So, when you look at these moving waves, you can freeze time (like taking a snapshot with a camera) and see its spatial features at a given time or stay at a fixed position in space and let the waves come at you as time passes to see the temporal features at a given position.

As we begin this Module, I want you to explore the characteristics of waves using PhET simulation, "Waves On A String." How do you generate waves in a string? How do you describe a wave, i.e., the parameters needed to define a wave fully? How do you make waves with different features?

BEFORE YOU START — PREDICT

You are about to drive one end of a string up and down, faster and faster. Before you touch a slider: when you **double the frequency**, does the wave travel along the string **faster**, and does the distance from one peak to the next get **longer**, **shorter** or **stay the same**? Write one sentence in your guided notes and commit to it. Item 19 is where you find out.

PERFORMANCE EXPECTATIONS

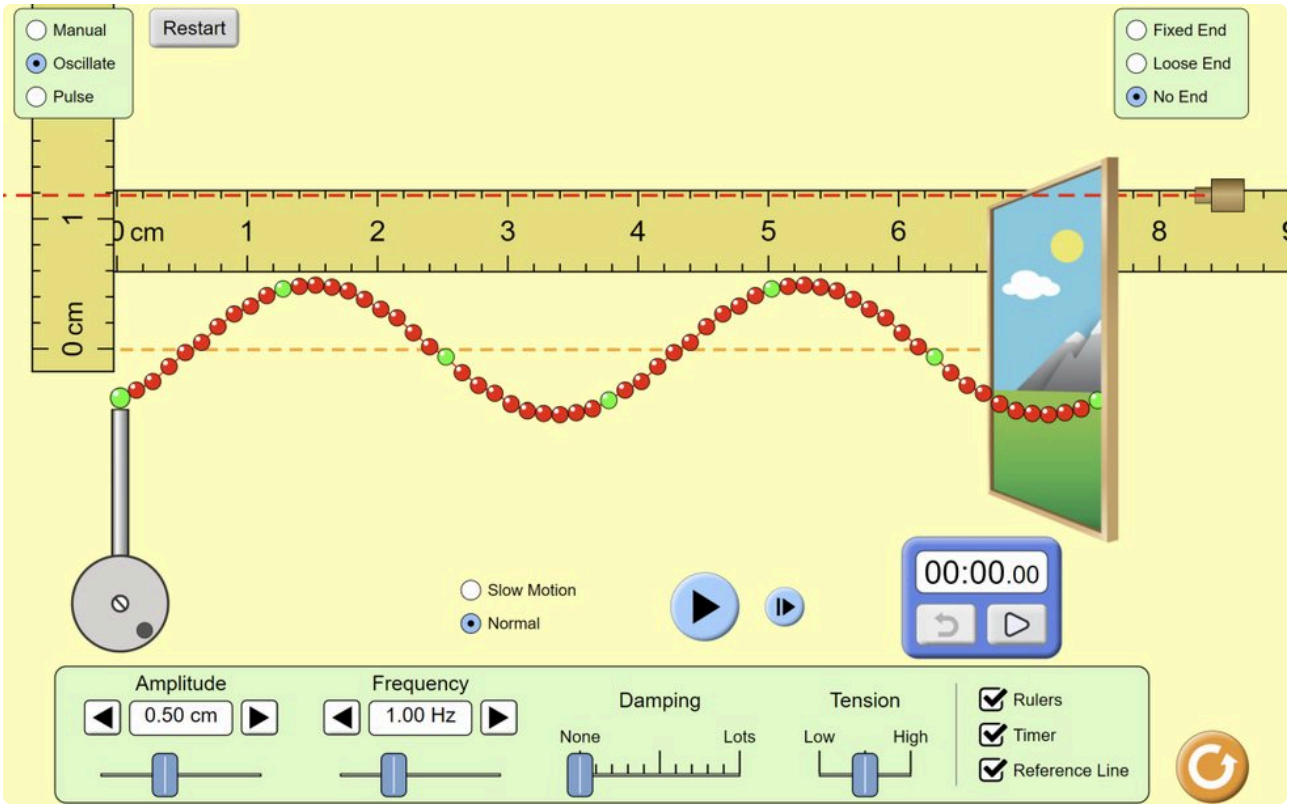
Define wave characteristics. Describe waves with amplitude, wavelength, and frequency. Generate waves with different amplitudes, wavelengths, and frequencies using appropriate controls in the simulation.

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Instructions and Questions

Open the simulation: phet.colorado.edu → "Waves on a String" (or follow item 1 and search for it).

1. Type "waves on a string phet" in the Google search bar. Run waves on a string PhET simulation.
2. Choose "Oscillate," "No End," and "Normal" options.
3. Slide Damping to None and Tension to Medium.
4. Check Rulers, Timer, and Reference Line.
5. Set Amplitude to 0.5 cm and Frequency to 1 Hz.
6. Check your screen looks like below (you can move the rulers).



Your screen after items 2–5. The horizontal ruler reads along the string, the vertical ruler reads displacement, and the yellow dashed line is the reference line.

7. Press the Play button to generate the moving wave. How is the moving wave generated?

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8. What does the Reference Line (the yellow dashed line) indicate?

9. Slide the amplitude back and forth, setting it back to 0.5 cm. What feature of the wave does it change? What is the value of the amplitude in reference to the Reference Line?

10. Freeze the wave in time by pressing the pause button. The wave is frozen in time, so you can easily observe its spatial features. A wave's wavelength is the peak-to-peak distance, trough-to-trough distance, or the length of one full repeat of the wave. What is the wavelength, λ (the Greek letter lambda)?

    λ = _____ cm

11. Press the Play button to let the wave move again. Which way is the wave moving?

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12. Imagine yourself standing at a 3 cm position on a horizontal ruler, facing the waves coming at you. Start the Timer as a peak of the wave passes you at the 3 cm position. Stop the Timer when you count the next ten peaks. How long did it take to count the 10 next peaks? Therefore, how long did it take to move from one peak to the next?

13. Repeat 12) with frequency set to 1.5 Hz, 2 Hz, 2.5 Hz, and 3 Hz. How long did it take to count 10 peaks after starting the Timer? Use the "Slow Motion" option if waves are moving too fast.

	Time for 10 peaks to pass	Time for 1 peak to pass
1.5 Hz		
2 Hz		
2.5 Hz		
3 Hz		

14. Determine the relationship between frequency (*f*, lowercase f) and the time taken for 1 peak to pass. (Hint: the unit Hz is equal to 1/s)

15. The time it takes to pass from one peak to the next is a wave's period, *T* (capital letter t). Note that the period is NOT the same as the wavelength! The period is the time it takes for one full repeat of the wave. What is the relationship between the frequency, *f*, and period, *T*? Don't forget units.

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16. The frequency, *f*, is the number of peaks you count in 1 second or the number of repeats (cycles) in 1 second. If you count 10 peaks in 5 seconds, how many repeats in 1 second? What is the frequency in Hz? Does this correspond to your table above?

17. If your wave has a frequency of 15 Hz, how many full repeats (cycles) are in 1 second? How long does it take for one full repeat (cycle)? What is the period of the wave? Don't forget units.

    $f =$ _____

    $T =$ _____

18. Moving waves clearly have speeds. The speed of a wave is simply how fast one full cycle runs past you if you were to stand at a fixed position and watch the wave move past you. Thus, the speed, v (m/s) is $v = \lambda/T$ (the wavelength (m) divided by period (s)). Show that the speed is frequency times wavelength.

19. Do you think that changing the frequency changes the speed of the wave? Change the frequency and measure the wavelength (freeze the wave and measure the wavelength). Fill in the table below and state your conclusion.

Frequency (Hz)	Wavelength (cm)	Speed (cm/s)
1		
2		
3		

Speed = frequency × wavelength

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20. Should changing the amplitude change the speed? (Hint: does the equation for the speed contain the amplitude?) Confirm your answer with simulation.

21. Does the tension affect the speed? If so, how?

22. What does damping do? What feature of the wave does it affect?

23. Choose “Pulse” option and press the green button to send a “pulse” along the string. Is it a wave? If so, how? If not, why not?

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Underline your answer for each situation

These four are about a **single pulse**. The hand made it and has stopped moving; the pulse is travelling to the right. The knot **A** is a marked bead of the rope.



One pulse, moving right. **A** is on the *front* face of the hump — the side nearer the direction of travel.

1. If you advance the movie one frame, the knot at point A would be
   a) in the same place b) higher c) lower d) to the right e) to the left

2. If the person generates a new pulse like the first but more quickly, the pulse would be
   a) same size b) wider c) narrower

3. If the person generates another pulse like the first but he moves his hand further, the pulse would be
   a) same size b) taller c) shorter

4. If the person generates another pulse like the first but the rope is tightened, the pulse will move
   a) at the same rate b) faster c) slower

These three are about a **continuous wave**, the hand still shaking.



A continuous train, moving right, with the hand still driving it.

5. If you advance the movie one frame, the knot at point A would be
   a) in the same place b) higher c) lower d) to the right e) to the left

6. If you advance the movie one frame, the pattern of the waves will be _____ relative to the hand.
   a) in the same place b) shifted right c) shifted left d) shifted up e) shifted down

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7. If the person starts over and moves his hand more quickly, the peaks of the waves will be
   a) the same distance apart b) further apart c) closer together

8. If you lower the frequency of a wave on a string you will
   a) lower its speed b) increase its wavelength c) lower its amplitude
   d) shorten its period

Looking ahead

You now have the three parameters the synopsis promised — amplitude, wavelength and frequency — and the two relations that tie them together, $T = 1/f$ and $v = \lambda f$. You also have the fact that does most of the work for the rest of the module: **the speed belongs to the medium, not to whoever is shaking it**. **Note 1a** fills in the vocabulary around those parameters; **Note 2** goes back to the one oscillating bead and asks what force makes it move that way, which is where the sines and cosines of the synopsis actually come from.

Where this sits. Everything here was measured, not derived — a ruler, a timer and a slider. That is deliberate: the definitions have to be yours before the algebra is worth doing. From Note 2 on, the same quantities arrive the other way round, out of Newton's second law.

BACK TO YOUR PREDICTION

Same speed, shorter wavelength. Doubling the frequency does *not* speed the wave up — the string sets the speed, through its tension — so the peaks simply arrive twice as often and sit half as far apart. If you predicted "faster", you were thinking of the bead's up-and-down motion, which really does get faster; that is not the speed of the wave along the string.

Check yourself

Four quick ones. Work them without looking back.

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